

GRADE: 7**TIME:** 40 minutes**MARKS: 30****EXAMINOR:** Mr LL Gaetsewe**MODERATOR:** Mr BG Matroos

This question paper consists of 4 pages.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of THREE SECTIONS: SECTIONS A, B AND C
2. Answer ALL the questions.
3. Leave a space between the subsections of the question paper.
4. Read ALL the questions carefully before you write the answers.
5. Number your answers exactly as they appear in the question paper.
6. Write neatly and legibly.
7. Sketches must be clear, neat and done in pencil.

SPECIFIC AIMS ASSESSED

- SA 1: TECHNOLOGICAL PROCESSES AND SKILLS (50%)
- SA 2: TECHNOLOGICAL KNOWLEDGE AND UNDERSTANDING (30%)
- SA 3: TECHNOLOGY, SOCIETY AND THE ENVIRONMENT (20%)

SECTION A	SPECIFIC AIM: 1	15 Marks
SECTION B	SPECIFIC AIM: 2	9 Marks
SECTION C	SPECIFIC AIM: 3	6 Marks

Mark distribution

Grade	Marks	Cognitive Levels			Weighting		
		Low order	Middle order	High order	S	K	V & A
		30%	40%	30%	50%	30%	20%
7	30	9	12	9	15	9	6

QUESTION 1

Read the scenario and complete the questions that follows:

Shack fires in South Africa.

Across South Africa, there are an estimated ten shack fires a day, often leaving residents dead and in some cases many thousands more homeless. In Cape Town, the 'fire capital of South Africa', the situation is particularly stark.

Matt Birkinshaw lists the key reasons for shack fires as lack of land, lack of housing, denial of access to electricity, inadequate water supply and inadequate emergency services.

The effect of such fires results in significant economic losses and a reduced quality of life for the affected community, and often the loss of lives. Fires kill thousands of people every year, with many more disabled or seriously injured, resulting in untold and often lifelong misery and entrenched poverty.

People tend to build their shacks too close to each other, which in turn increase the risk of informal settlement fire. Due to the lack of basic services, like electricity, people are compelled to use alternative energy sources.

Safety hints and tips to keep in mind to prevent shack fires:

- Never cook near materials that can burn.
- Make sure that all stoves are on a stable surface to prevent them from falling over.
- Make sure all candles are placed in a safe, non-combustible candle holder.



Figure1: Shacks burning in Langa

- 1.1 Point out how many shack fires are daily experience in South Africa? (1)
- 1.2 Cape Town is referred to as the “fire capital of South Africa”. Discuss your understanding of the phrase. (1)

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| 1.3 | Relate TWO reasons from the scenario the cause of shack fires? | (2) |
| 1.4 | Name TWO alternative energy sources that can easily replace electricity? | (2) |
| 1.5 | Recommend TWO possible ways to prevent shack fires. | (2) |
| 1.6 | Write a design brief for an appropriate design with specifications for producing a suitable textile/ fabric/ material to use in making an emergency shelter. (Indicate the material to be used, where will it be placed and when it must be use) | (3) |
| 1.7 | Design with labelling a possible idea for an emergency shelter that can easily be transported. | (4) |

SECTION B: TECHNOLOGICAL KNOWLEDGE AND UNDERSTANDING

[9]

QUESTION 2

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|-----|---|-----|
| 2.1 | Differentiate between a magnet and an electro magnet and give an example of a permanent magnet? | (2) |
| 2.2 | Distinguish between ferrous and non-ferrous metal and give an example of each metal? | (2) |
| 2.3 | Describe how to make an electromagnet? | (2) |



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| 2.4 | Draw an electromagnet showing the magnetic field with two cells connected in parallel. 2 | (3) |
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SECTION C: TECHNOLOGY, SOCIETY AND THE ENVIRONMENT

[6]

QUESTION 3

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|-----|---|-----|
| 3.1 | List TWO negative effects that STAGE 4 load shedding have on the school environment. | (2) |
| 3.2 | Describe the effects of informal settlements on the environment? | (2) |
| 3.3 | Discuss the ingredients of a tasty, nutritious meal which you would recommend for a refugee camp. | (2) |

TOTAL 30

THE END